DO NOW — RECALL

Noughts & Crosses · Part 4 — Complete and test

Name: _____ Class: _____ Date: _____

1. What does MakeMove() return after a successful move?

2. For a 3 × 3 C# array, what does board.GetLength(0) return?

3. In Triple(7) , identify the argument. In Triple(int number) , identify the parameter.

4. A 3 × 3 game board is displayed using repeated row-printing statements. Explain how nested loops improve the code and what each loop controls.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

true.

2. Model answer

3 — the number of rows.

3. Model answer

Argument: 7. Parameter: number.

4. Model answer

An outer loop visits rows 0–2; an inner loop visits columns 0–2. The same operation is reused instead of duplicated, making the code easier to maintain.

By the end, you can…

• complete HasWon() for rows, columns and diagonals;

• detect a draw using BoardFull() ;

• finish, test and debug a complete playable game.

Vocabulary — know these terms

Recall the term — Boolean

Recall the term — Condition

Recall the term — State

Recall the term — Game loop

Recall the term — Test case

Recall the term — Trace

Recall the term — Debug

Recall the term — Draw

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

Complete the condition
CODE:
```
if (board[0,0] == player &&
board[1,1] == player &&
board[2,2] == player)
{
return true;
}
```

Guided 2 — CHECK

Build the Boolean test

Guided 3 — CHECK

Complete the sequence

Guided 4 — CHECK

Predict the functions

NEW LEARNING

Read, annotate and explain

Eight ways to win

3 rows + 3 columns + 2 diagonals = 8 win lines. To finish the game, add HasWon(player) , BoardFull() and the game loop. Test every line, not just the top row.

HasWon() checks 8 lines

Each winning line needs three comparisons joined with && . The possible lines are joined with || .

BoardFull() looks for any free square

If even one square is free, the board is not full.

Finish Main()

Part 3 already handled valid moves and switching. Today we add the stopping conditions.

Can two pupils actually play it?

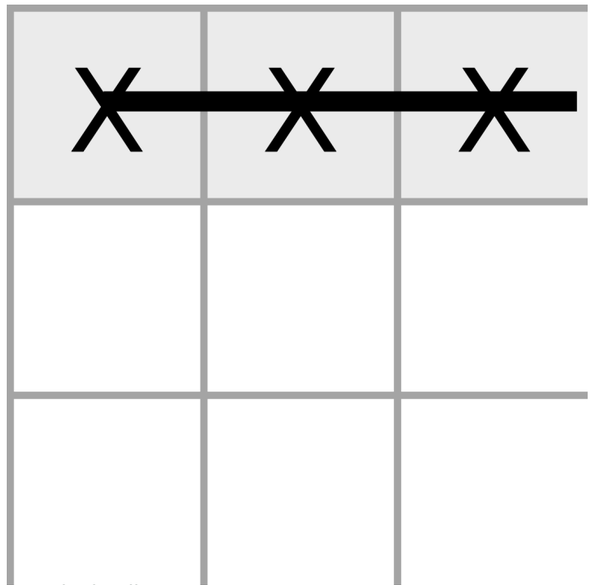
After completing the code, play against a partner and compare expected with actual results. Use planned board positions to test all eight winning lines.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Eight ways to win



Row 1: three X marks in one winning line

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

After a valid move, the program should check HasWon() before BoardFull() .

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Complete the game loop
Draw the board, ask the current player for a move, call MakeMove() , then check for a win or draw.
Switch player only after a valid move that did not end the game.

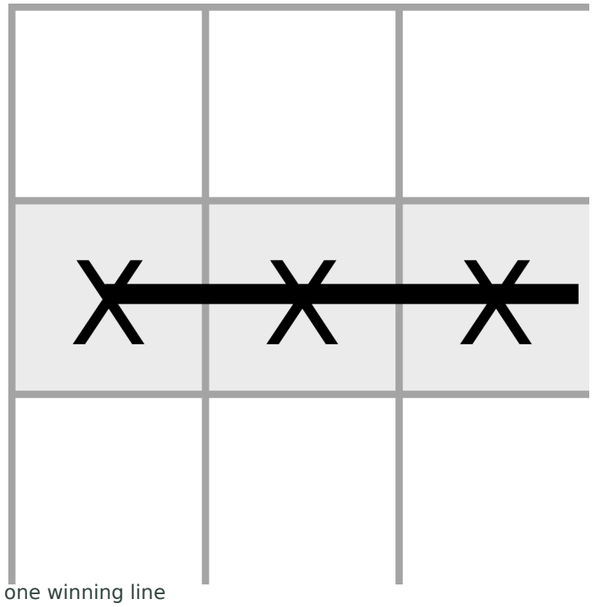
Question 4

Partner challenge: play and test
Pair up: one pupil plays X and the other O. Run your own C# program, make moves 1–9 and swap who goes first for a second complete game.
Evidence: for each targeted test, record the input or board state, expected result, actual result and pass/fail. Fix and retest any failure. Test all eight winning lines using prepared moves or board states; ordinary matches alone may not cover them.
If your online C# runner does not support repeated keyboard input, run the program in a local C# console or use planned input sequences for testing.

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Eight ways to win

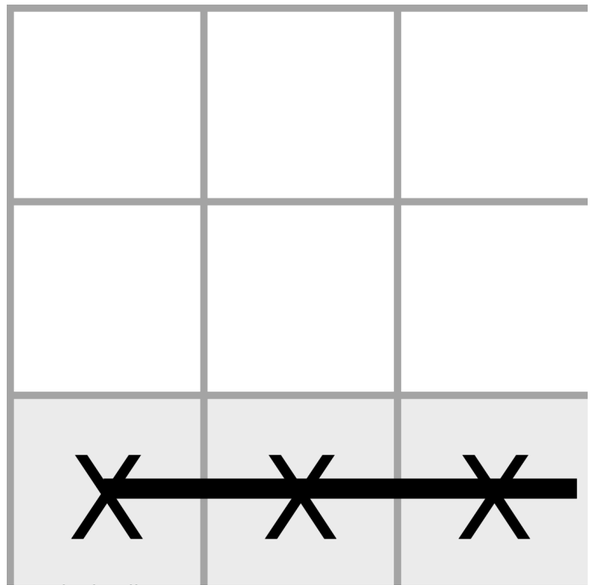


Row 2: three X marks in one winning line

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Eight ways to win



Row 3: three X marks in one winning line

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Finish all 8 winning lines
Continue your Lesson 015 program.
Complete HasWon(string player) so it detects all 3 rows, all 3 columns and both diagonals.

Question 2

Detect a draw
Create BoardFull() .
Return false as soon as a square contains neither X nor O. Return true only if every square is occupied.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY: Complete one diagonal

Player X wins on the main diagonal when all three diagonal positions contain X.

```
if (board[0,0] == ? &&
    board[1,1] == ? &&
    board[2,2] == ?)
{
    return true;
}
```

Method / working:

Guided 2 — TRY: Spot a free square

A square is free when it contains neither X nor O.

Method / working:

Guided 3 — TRY: Put the actions in order

A valid move has just been made. What should happen next?

Method / working:

Guided 4 — TRY: Choose the test

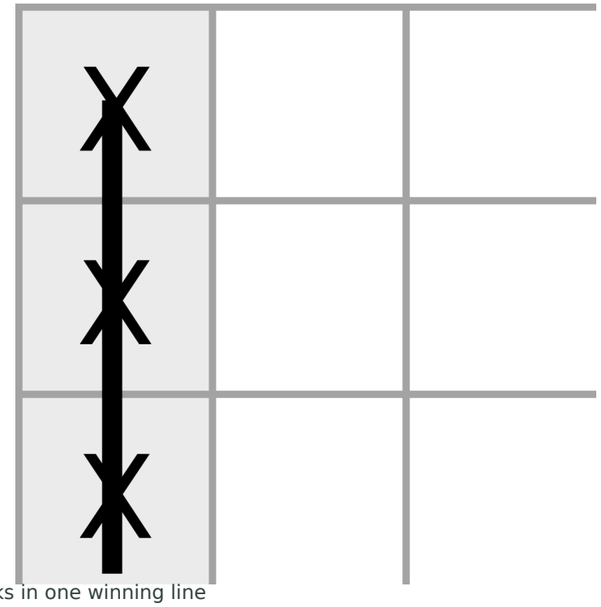
No winning line exists and every square is occupied.

Method / working:

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Eight ways to win

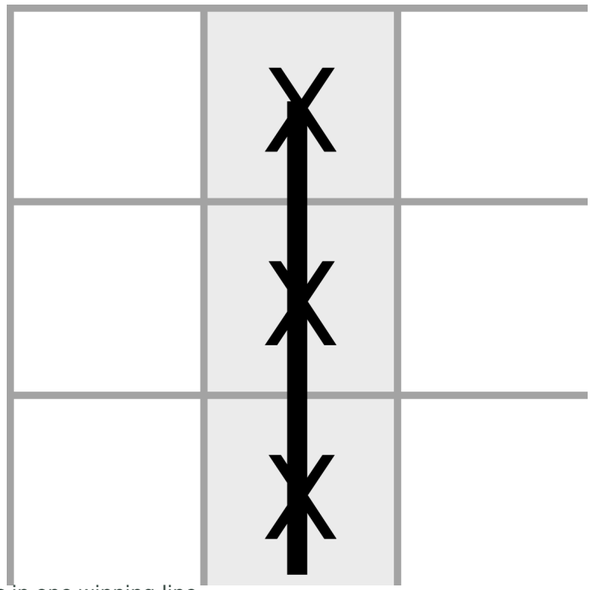


Column 1: three X marks in one winning line

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Eight ways to win



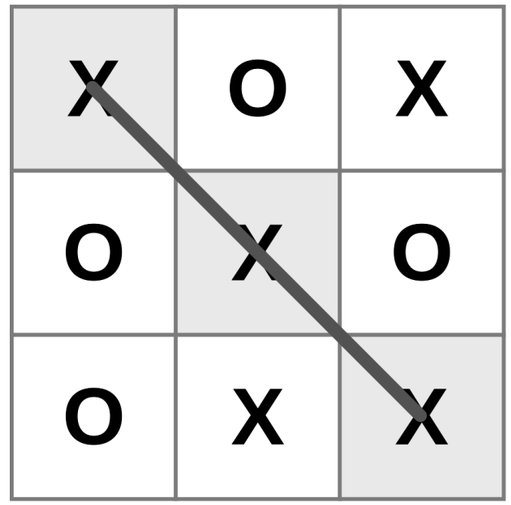
Column 2: three X marks in one winning line

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Build BoardFull()

Return true only when every square contains X or O.



Full board AND X has won

Full Noughts and Crosses board, with X winning on the top-left to bottom-right diagonal

I DO — WORKED EXAMPLE

Follow the worked method

Build BoardFull()

Use nested loops

```
for (int row = 0; row < board.GetLength(0); row++)
{
    for (int column = 0; column < board.GetLength(1); column++)
    {
    }
}
```

One free square means not full

```
if (board[row, column] != "X" &&
    board[row, column] != "O")
{
    return false;
}
```

The function can stop immediately when it finds a free square.

No free square was found

```
return true;
```

This line is reached only after every position has been checked.

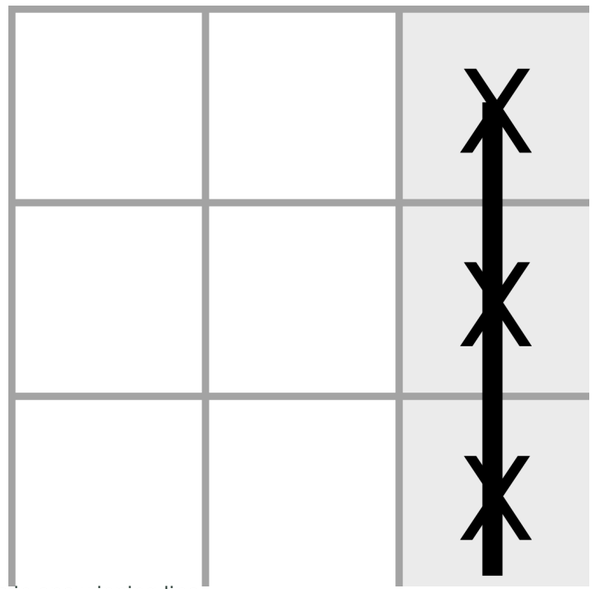
Check draw after checking win

```
if (HasWon(currentPlayer))
{
    // win
}
else if (BoardFull())
{
    // draw
}
```


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Eight ways to win

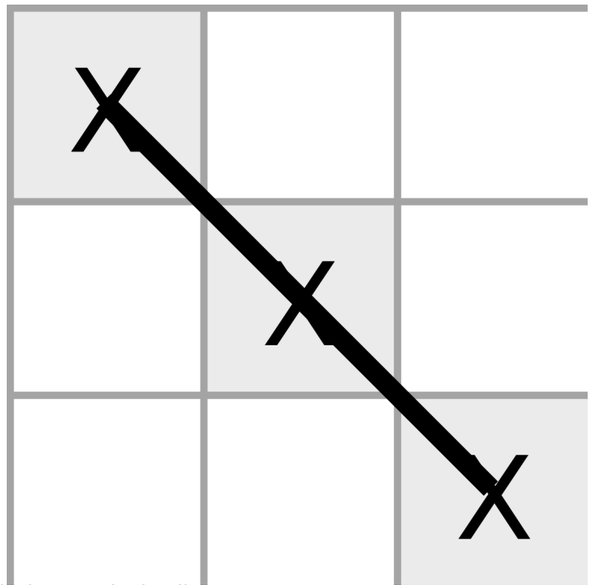


Column 3: three X marks in one winning line

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Eight ways to win



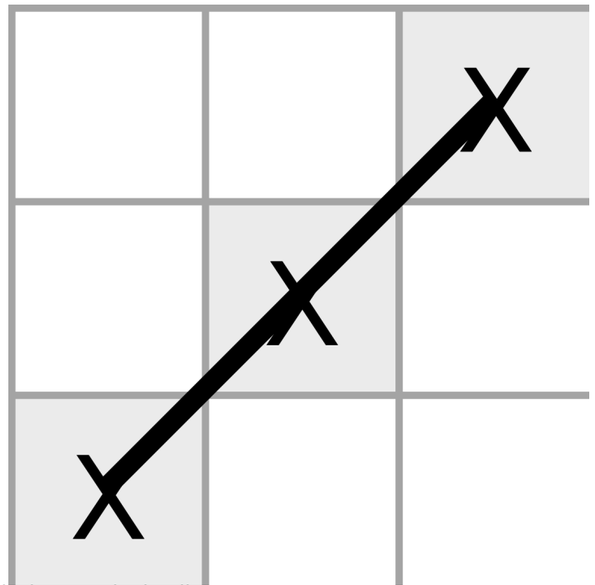
Diagonal ↘: three X marks in one winning line

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Eight ways to win



Diagonal ↙: three X marks in one winning line